



Match the Stack

Quick facilitation notes & answer key

Goal

Children match a block program to the result it produces and, given a wanted result, choose the program that produces it. Running forward (program → box) and reasoning backward (wanted box → program) is an early form of abstraction: a program stands for the outcome it makes.

Kindergarten frame (Ontario)

Problem Solving and Innovating — children relate a set of instructions to its outcome and select instructions to meet a goal.

No reading required of the child: read each prompt aloud; the child circles a box card or circles P / Q.

Materials

The printed worksheet and a pencil. Optional: floor boxes 1-5 with a star on box 4 so children can run P and Q and see which lands on the star.

Run it (about 15 min)

1. MODEL: run Program A (➡ ➡) from box 1 together — box 2, box 3. 'Program A matches box 3.'
2. GUIDED: Exercise 1 — children circle the box-3 card. Exercise 2 — run P (➡ ➡ ➡ ➡ → box 4, the star) and Q (➡ → box 2); P lands on the star.
3. YOU TRY: Exercise 3 — same two programs, but now we want box 2; Q is the match.
4. Connect: 'A program stands for where it sends Bit. Pick the program that matches the box you want.'

Answer key (modelled by the run-gate)

Ex 1 — Program A (➡ ➡) from box 1 → box 3. Circle box 3.

Ex 2 — Program P (➡ ➡ ➡ ➡) → box 4, the star. Circle P.

Ex 3 — Program Q (➡) → box 2. Circle Q.

Watch for & differentiate

Common slip in Ex 2-3: children pick by stack height alone — have them actually run each program to the box before circling.

Simplify: do Exercise 1 and 2 only (forward match, then find the star).

Extend: ask which program would match box 5 (it would need four ➡ blocks), or have a child invent a program for box 3.

Movement option: two children each 'become' Program P and Program Q and walk their steps; the class names the box each one matches.

Success looks like

The child circles box 3 for Program A, Program P for the star, and Program Q for box 2 — running each program to confirm the match.